Computer Science & DA

Warehousing

Lecture - 02

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Recap of Previous Lecture

Topic Introduction

Topic Steps in Datawarehouse

Topics to be Covered

- Topic: ETL
- Topic: ELT Based Warehousing
- Topic: Data Mart
- Topic: Datawarehouse Schemas

Topic : Data Warehousing

Topic : ETL (Extract, Transform, Load)

Three-phase process where data is extracted, transformed and loaded into an output data container.

Topic : ETL (Extract, Transform, Load)

Extract:

- Involves extracting data from homogeneous or heterogeneous sources
- An intrinsic part of the extraction involves data validation

Topic : ETL (Extract, Transform, Load)

Transform:

A series of rules or functions are applied to the extracted data in order to prepare it for loading

Topic : ETL (Extract, Transform, Load)

Transform:
A series of rules or functions are applied to the extracted data in order to prepare it for loading

Cleansing:
Aims to pass only "proper" data to the target

Topic : Transform

- Selecting only certain columns to load
- Translating coded values (e.g., if the source system codes male as "1" and female as "2", but the warehouse codes male as "M" and female as "F")
- Encoding free-form values (e.g., mapping "Male" to "M")
- Deriving a new calculated value (e.g., sale_amount = qty * unit_price)
- Sorting or ordering the data based on a list of columns to improve search performance
- Joining data from multiple sources (e.g., lookup, merge) and deduplicating the data

Reliance digital ⇒ Customers

Reliance fresh ⇒ Customers

D.W.

Topic : Transform

- Aggregating (for example, summarizing multiple rows of data — total sales for each store, and for each region, etc.)
- Generating surrogate-key values
- Transposing or pivoting (turning multiple columns into multiple rows or vice versa)
- Splitting a column into multiple columns (e.g., converting a comma-separated list, specified as a string in one column, into individual values in different columns)
- Applying any form of data validation; failed validation may result in a full rejection of the data, partial rejection, or no rejection at all.

Database

student

rno.	name	dob	fname	city
1				
2				
3				
4				
5				
6				

key

primary key

Datawarehouse

seq. no.	rno		
1	1	—	—
2	1	—	—
3	1	—	—
4	1	—	—
5			
6			

surrogate key

Topic : ETL (Extract, Transform, Load)

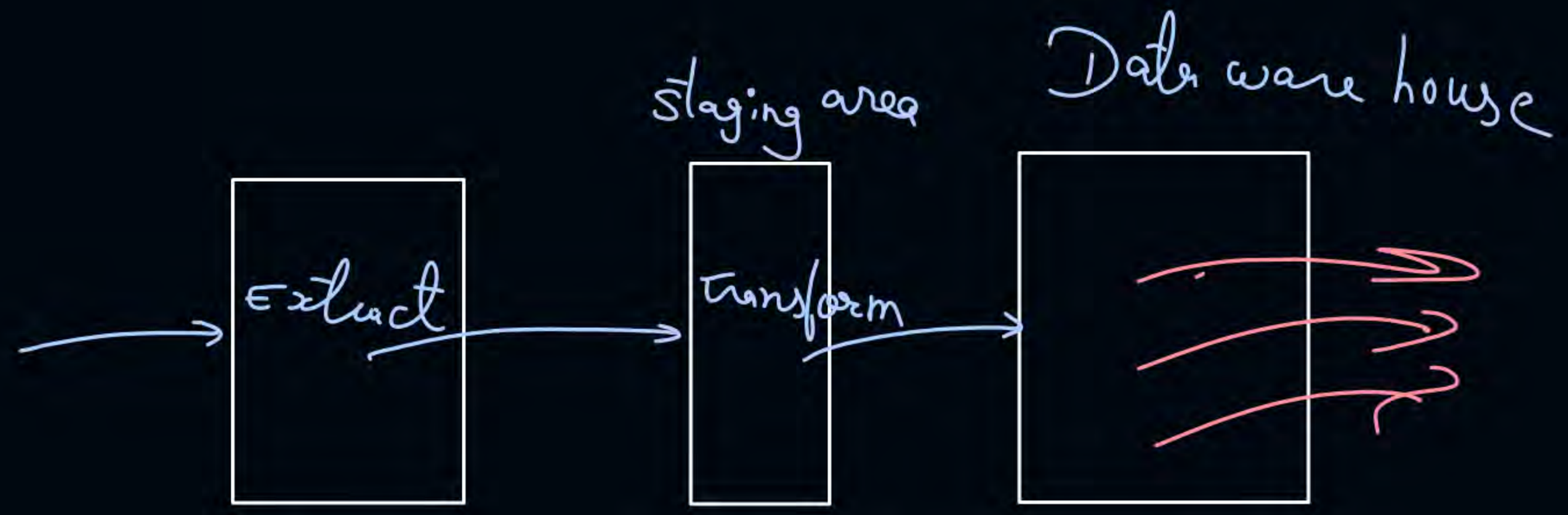
Load:

This phase loads the data into the end target

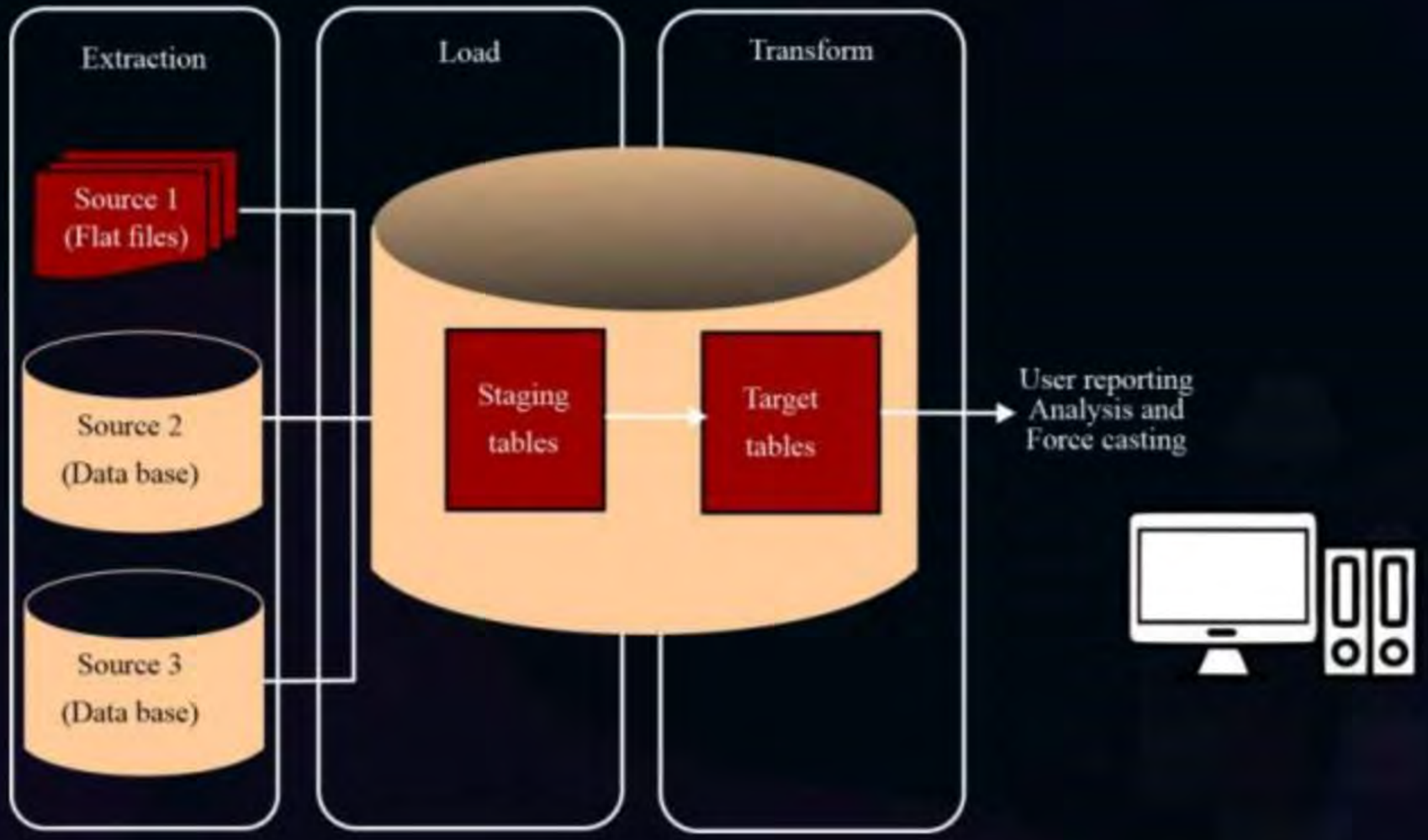
Topic : Data Warehousing

ETL Based: Extract, Transform, Load ⇒ Corona
Uses Staging

ELT Based: Extract, Load, Transform ⇒ normal
gets rid of a separate ETL tool for data transformation. Instead, it maintains a staging area inside the data warehouse itself.


Staging area
Data ware house
Extract
Transform

Topic : ELT Based Data Warehousing

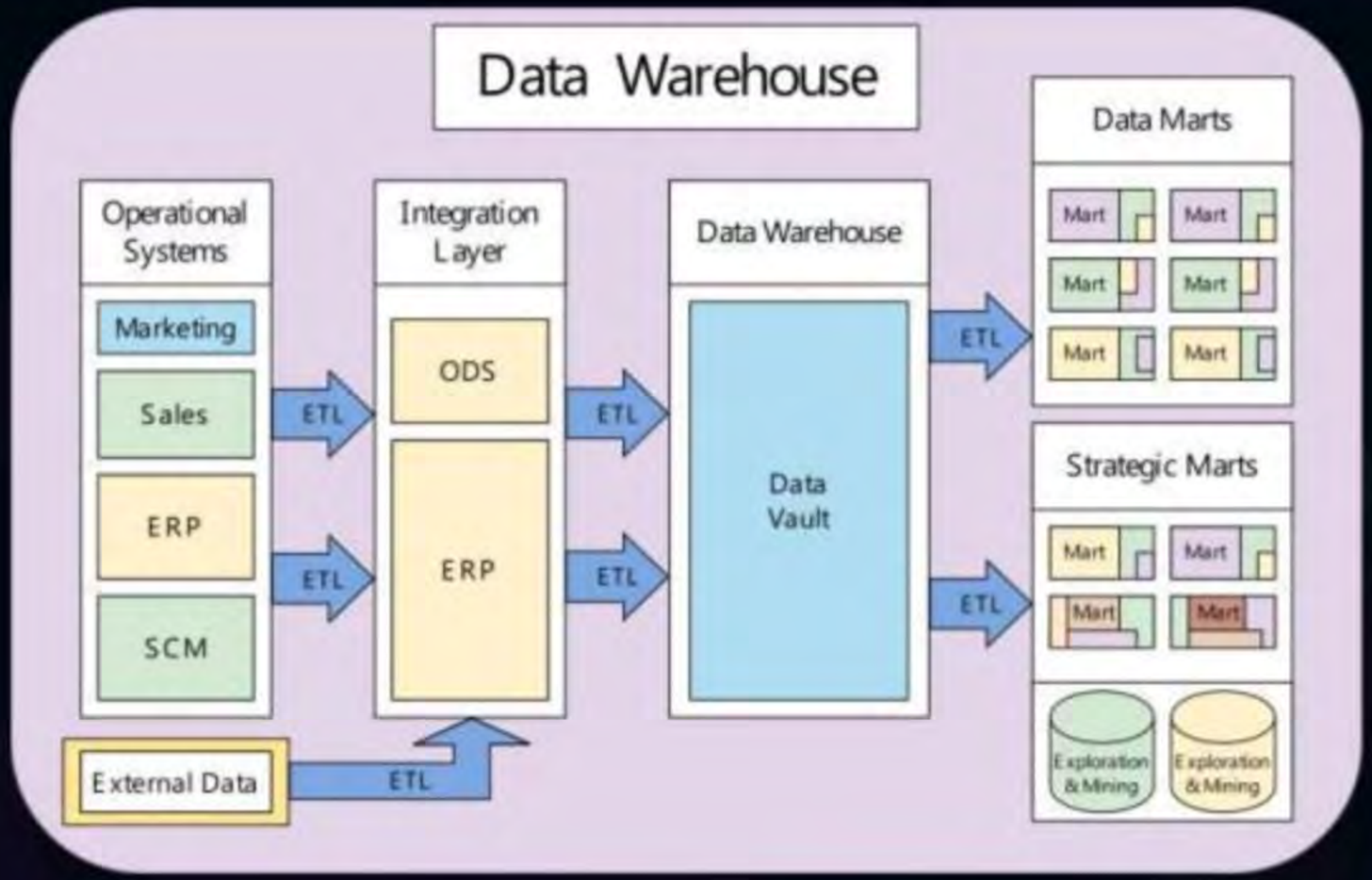


Topic : Data Mart

- Simple form of a data warehouse, focused on a single subject (or functional area), hence they draw data from a limited number of sources such as sales, finance or marketing.

- Data marts are often built and controlled by a single department within an organization.

Topic : Data Mart



Topic : Data Warehousing vs Data Mart

Data warehouse:

- Holds multiple subject areas
- Holds very detailed information
- Works to integrate all data sources
- Does not necessarily use a dimensional model but feeds dimensional models

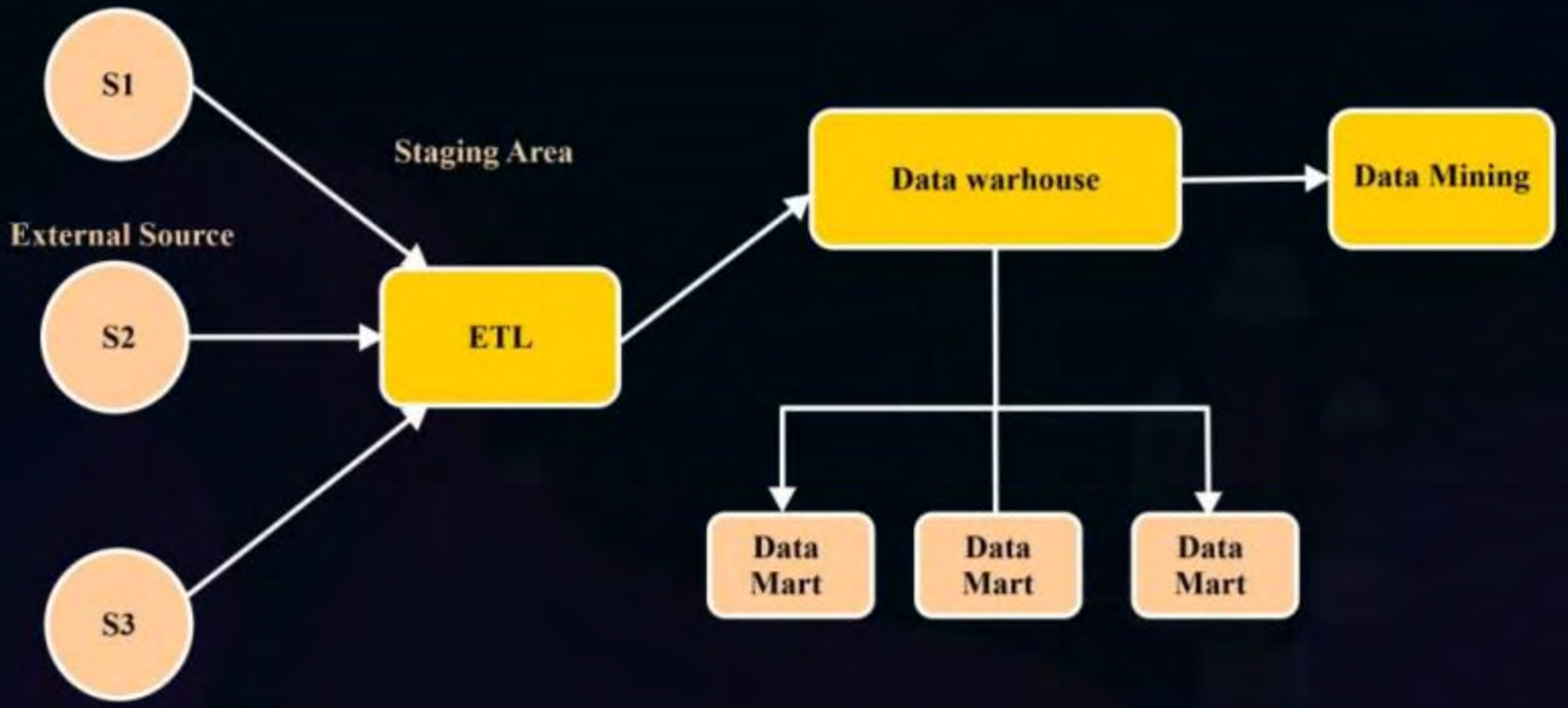
Data mart:

- Often holds only one subject area- for example, Finance, or Sales
- May hold more summarized data (although it may hold full detail)
- Concentrates on integrating information from a given subject area or set of source systems
- Is built focused on a dimensional model using a star schema.

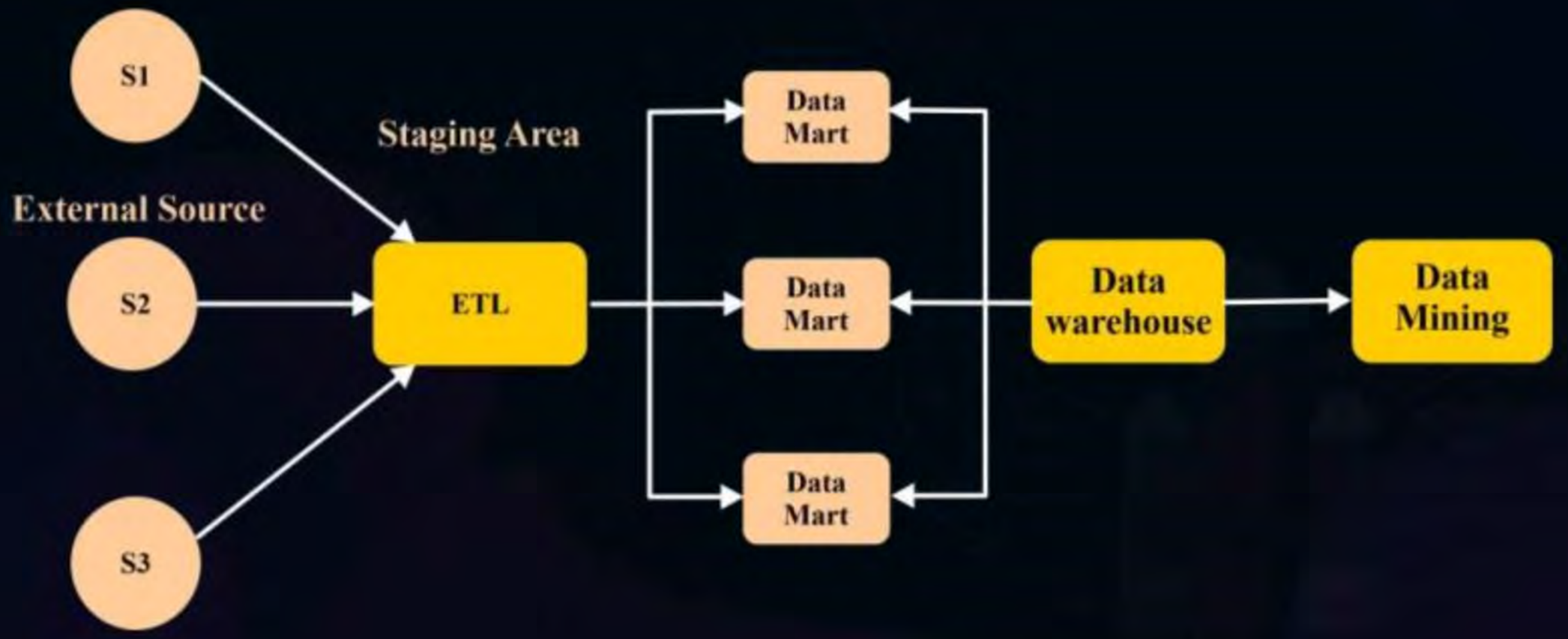
Topic : Data Warehousing Architecture

- Top-down approach
- Bottom-up approach

Topic : Top-down approach



Topic : Bottom-up approach



Topic : Online Transaction Processing (OLTP)

Type of data processing that consists of executing a number of transactions occurring concurrently. Done on databases

Examples:

Online banking, shopping, order entry, or sending text messages

Topic : Online Analytical Processing (OLAP)

Software for performing multidimensional analysis at high speeds on large volumes of data from a data warehouse, data mart, or some other unified, centralized data store

Topic : OLTP vs OLAP

OLTP systems	OLAP systems
Enable the real-time execution of large numbers of database transactions by large numbers of people	Usually involve querying many records (even all records) in a ~~database~~ data warehouse for analytical purposes
Require lightning-fast response times	Require response times that are orders of magnitude slower than those required by OLTP
Modify small amounts of data frequently and usually involve a balance of reads and writes	Do not modify data at all; workloads are usually read-intensive
Use indexed data to improve response times	Store data in columnar format to allow easy access to large numbers of records
Require frequent or concurrent database backups	Require far less frequent database backup
Require relatively little storage space	Typically have significant storage space requirements, because they store large amounts of historical data
Usually run simple queries involving just one or a few records	Run complex queries involving large numbers of records

2 mins Summary

Topic ETL

Topic ELT Based Warehousing

Topic Data Mart

Topic Datawarehouse Schemas

Happy Learning

THANK - YOU